

Acreage Estimation and Uncertainty Quantification for New York State

MPS in Applied Statistics and Data Science Capstone Project

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Project Overview

- Motivation & objective
- Data & feature engineering
- Modeling & Evaluation
- Summary
- Area for improvement



Motivation

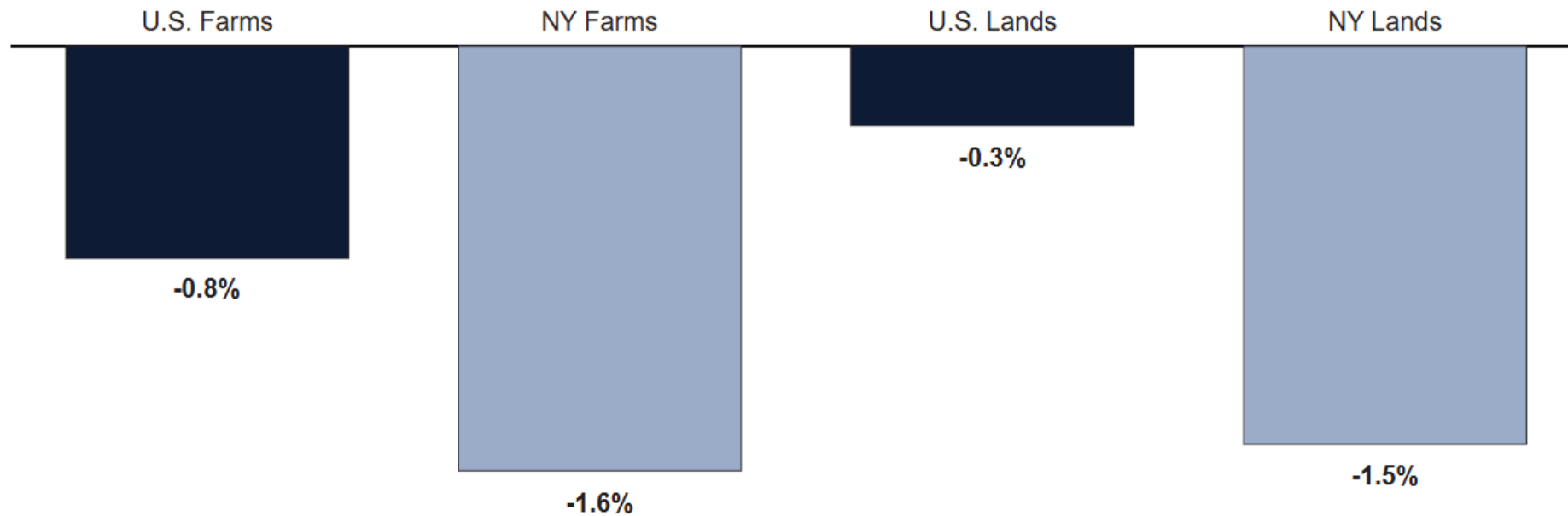
- US farmland in New York State has been declining over time
- Farm operations face rising costs and financial pressure
- Uncertainty in agricultural land use affects future food supply
- Forecasting acreage helps improve planning and decision-making



Motivation

Decline in farmland area

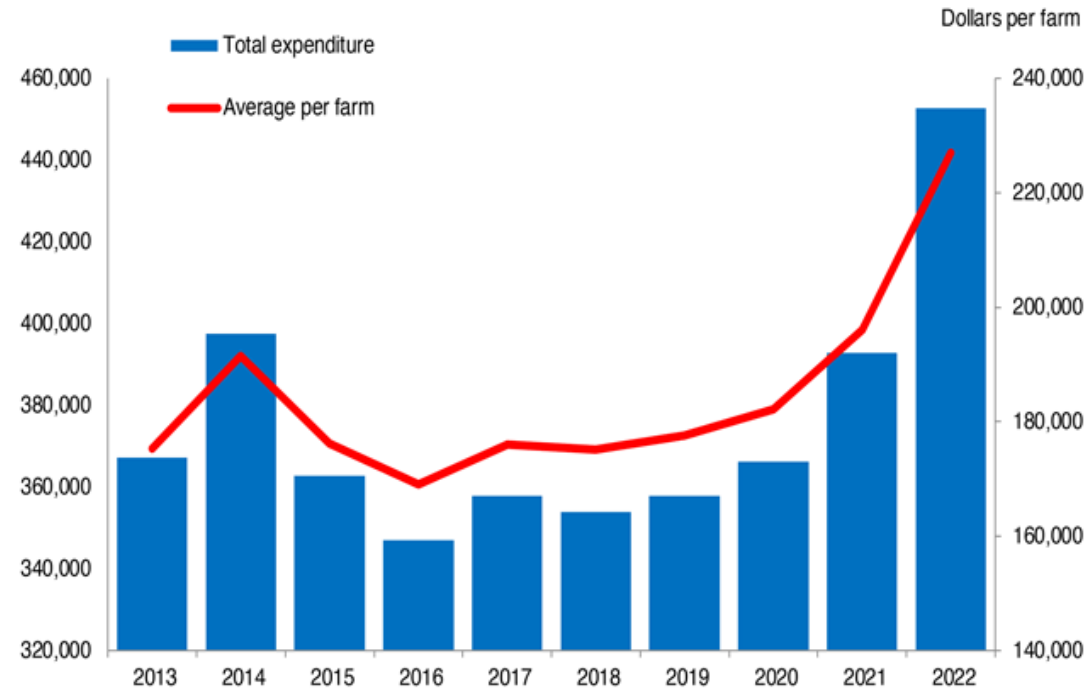
Percent Change in U.S. and New York Farms and Land in Farms, 2024-2025



Motivation

Increasing farm production expenditures

Farm Production Expenditures, Total, and Average Per Farm by Year – United States: 2013-2022



Objective

- Utilize historical crop spatial, weather and soil data
- Develop a predictive model to forecast crop-specific acreage at the county level in New York State
- Predict future acreage trends across crops and regions
- Support data-driven decision making for farmers and policymakers



Data - CSB (Crop Sequence Boundaries)

- A satellite-derived dataset across the contiguous U.S. on agriculture
- Year: 2008-2024 (Annual)
- Column: (polygon level)
 - Field-level crop rotations
 - Crop type
 - Acreage (in CDL)
 - Geometric information (county, state, longitude and latitude)
- The data is grouped and merged in a wide format. (1,553,005 polygons)
- The coding logic of the item is obtained from USDA website.

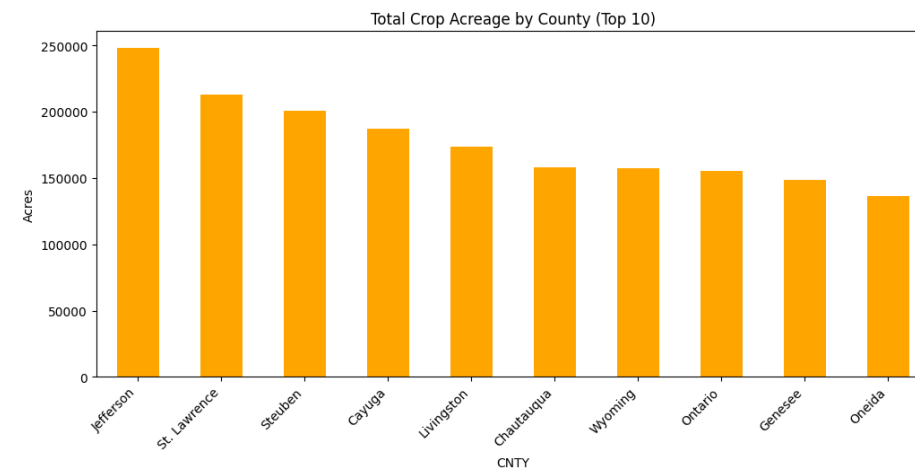
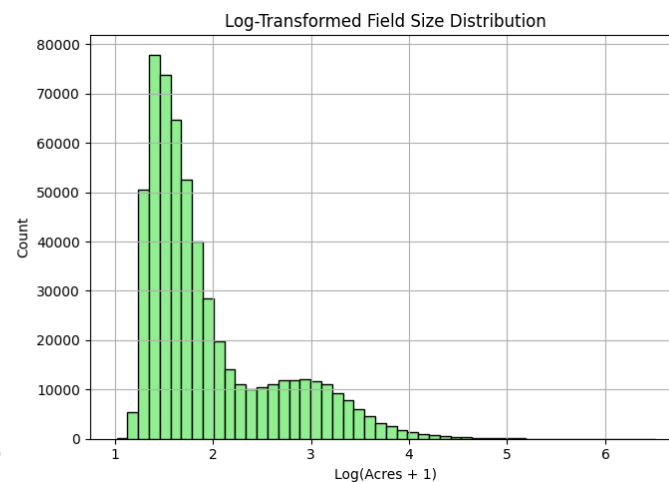
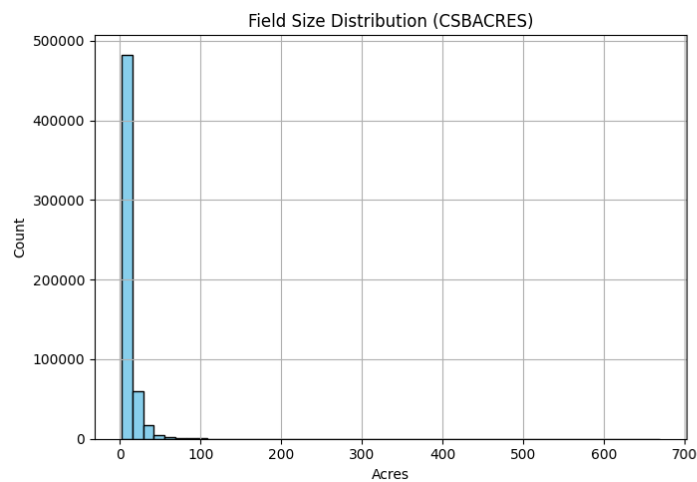
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DataFrame Info:
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Index: 569138 entries, 1095234 to 19696982
Data columns (total 21 columns):
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1   CSBYEARS               569138 non-null  str
2   CSBACRES               569138 non-null  float64
3   CDL2009               569138 non-null  int32
4   CDL2010               569138 non-null  int32
5   CDL2011               569138 non-null  int32
6   CDL2012               569138 non-null  int32
7   CDL2013               569138 non-null  int32
8   CDL2014               569138 non-null  int32
9   CDL2015               569138 non-null  int32
10  CDL2016               569138 non-null  int32
11  STATEFIPS             569138 non-null  str
12  STATEASD              569138 non-null  str
13  ASD                   569138 non-null  str
14  CNTY                  569138 non-null  str
15  CNTYFIPS              569138 non-null  str
16  INSIDE_X              569138 non-null  float64
17  INSIDE_Y              569138 non-null  float64
18  Shape_Length          569138 non-null  float64
19  Shape_Area            569138 non-null  float64
20  geometry               569138 non-null  object
dtypes: float64(5), int32(8), object(1), str(7)
memory usage: 97.5+ MB
```



Data - Acreage Distribution

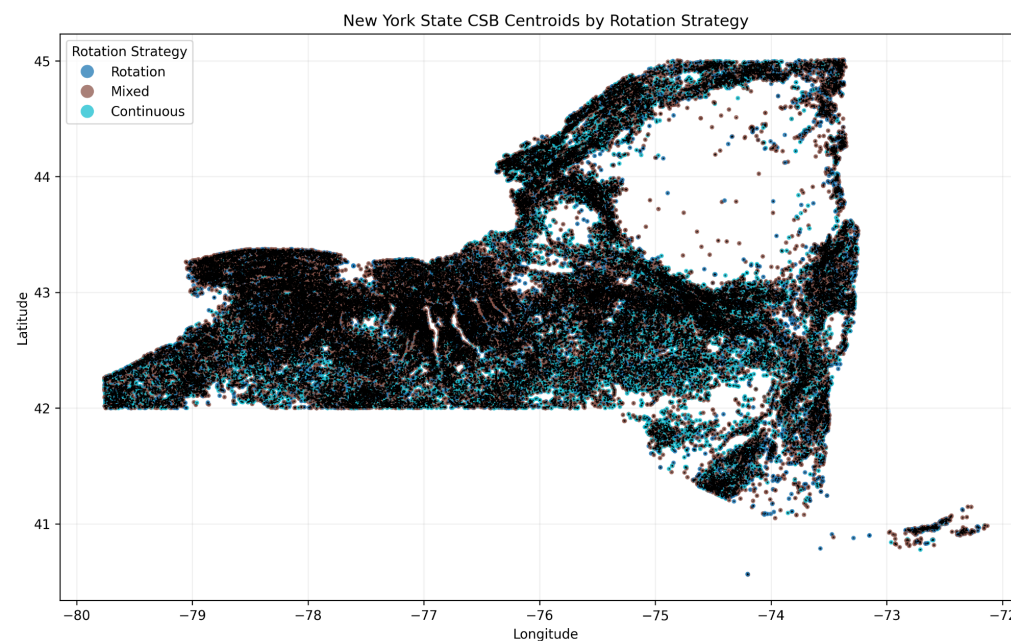
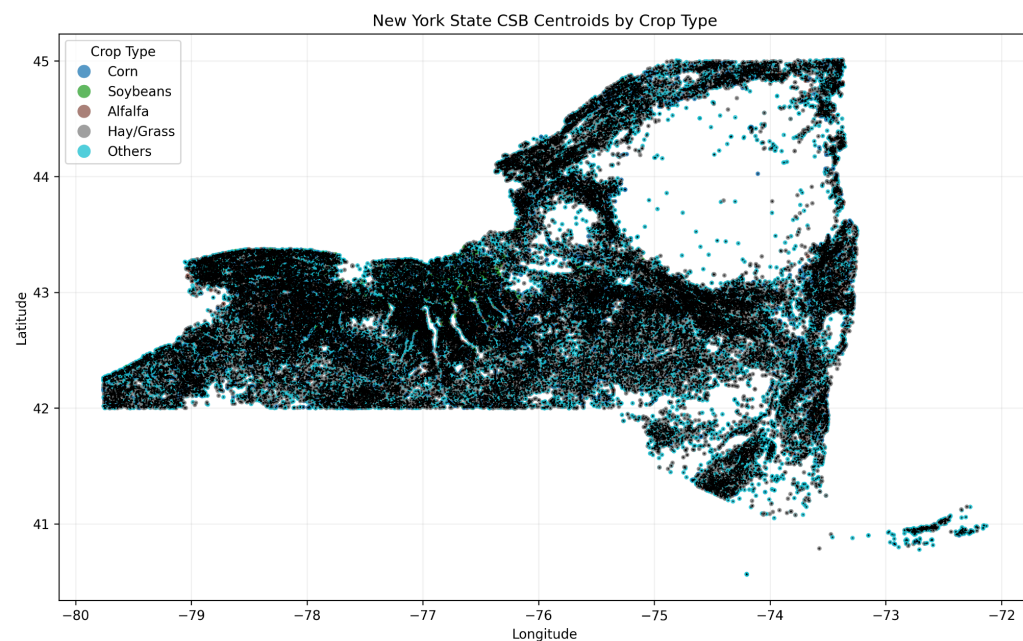
- Heavy right skewed
- Fat Right tail
- Balanced across County

Skewness: raw: 6.9 vs log-transformed: 1.4



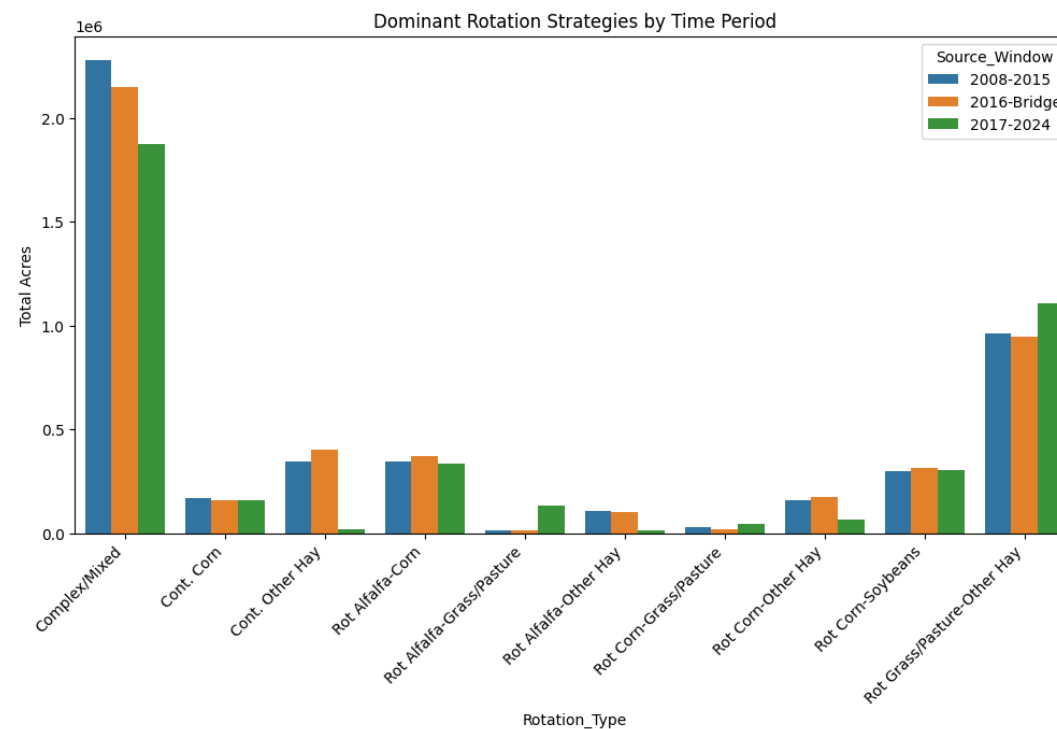
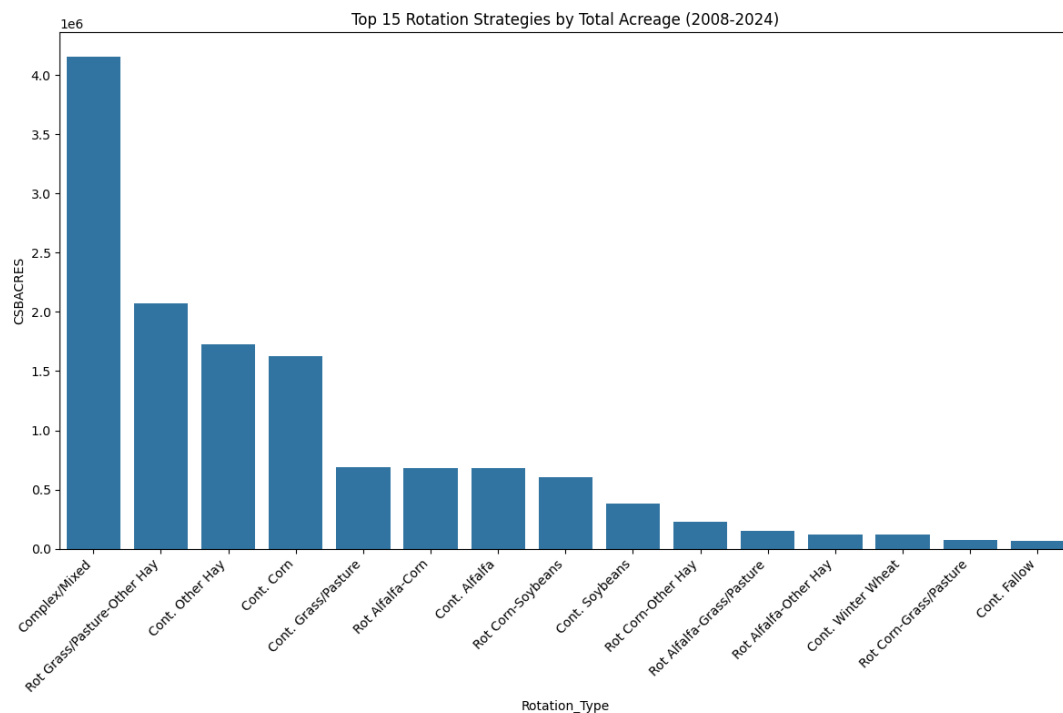
Data - Geometric Data

- The CSB dataset captured the polygon data by specifying their longitude and latitude.
- The data is in parquet format to store the geometric label



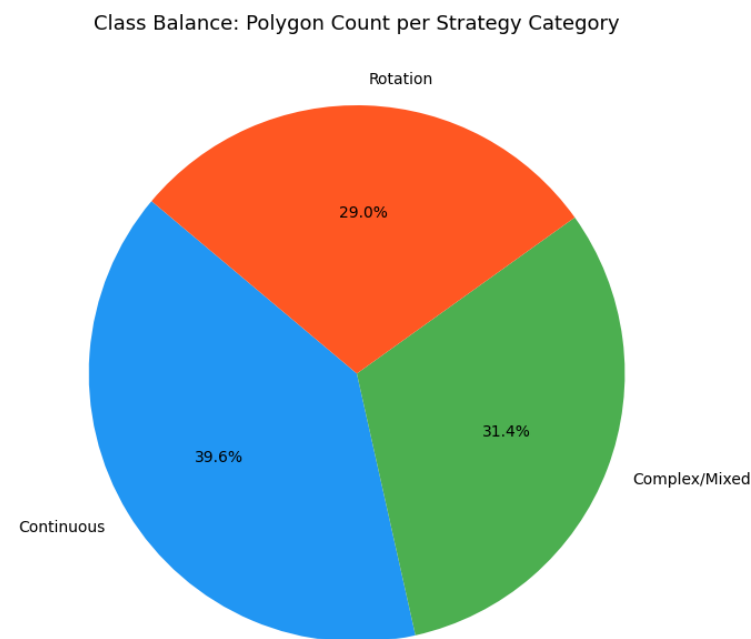
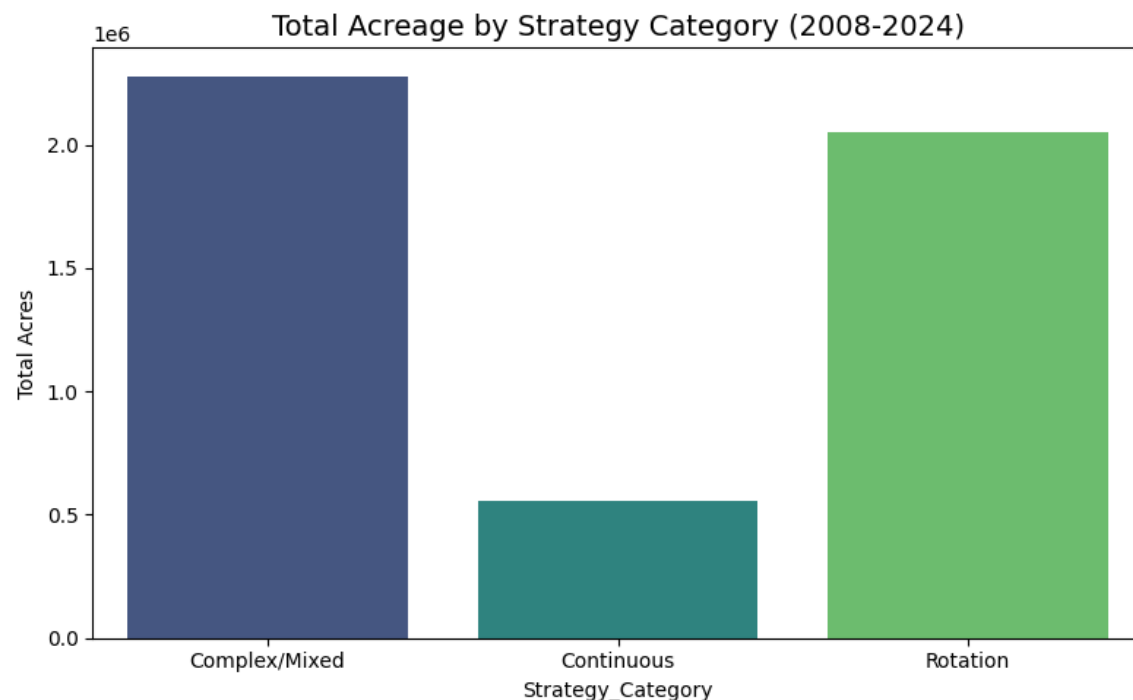
Data - Rotational Strategy

- The rotational strategies spanned large dimensions of column
- We decided to group the strategy type to simply the model and balance the data
- Potential issue: smoothed out the trend and specific strategy feature



Data - Class Imbalance

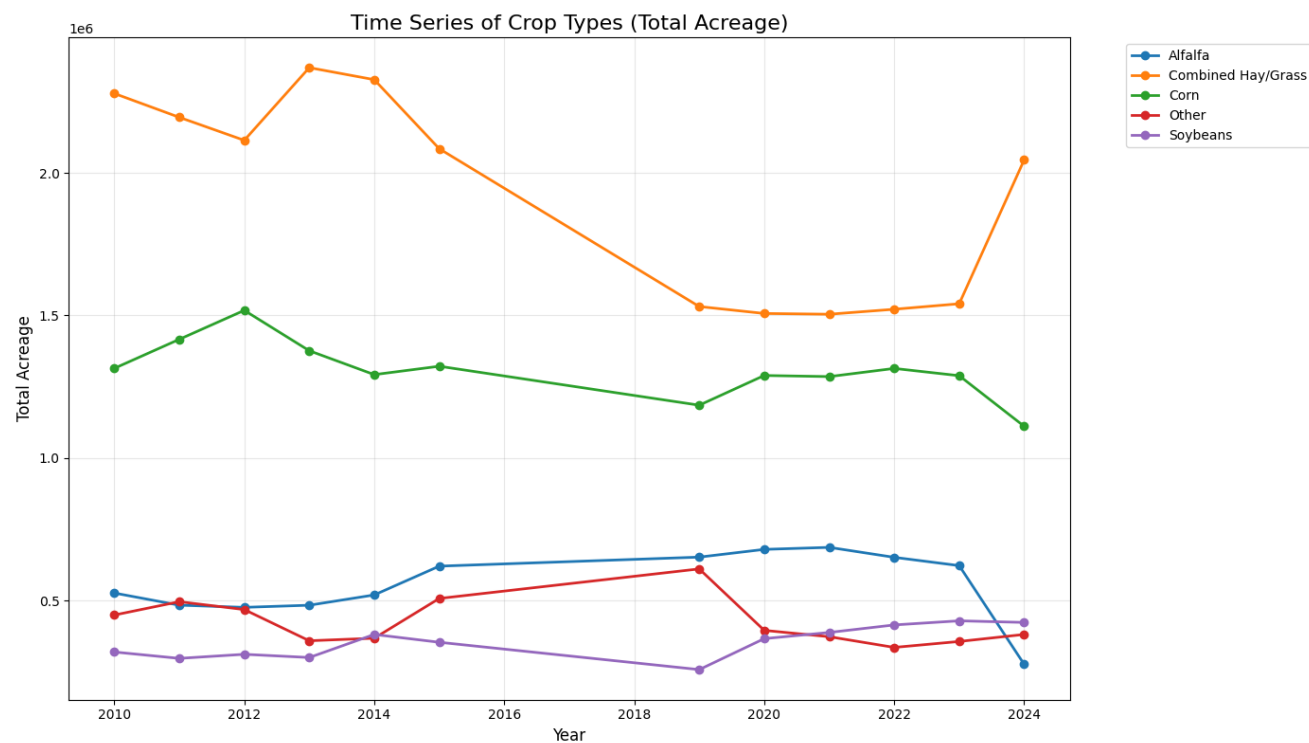
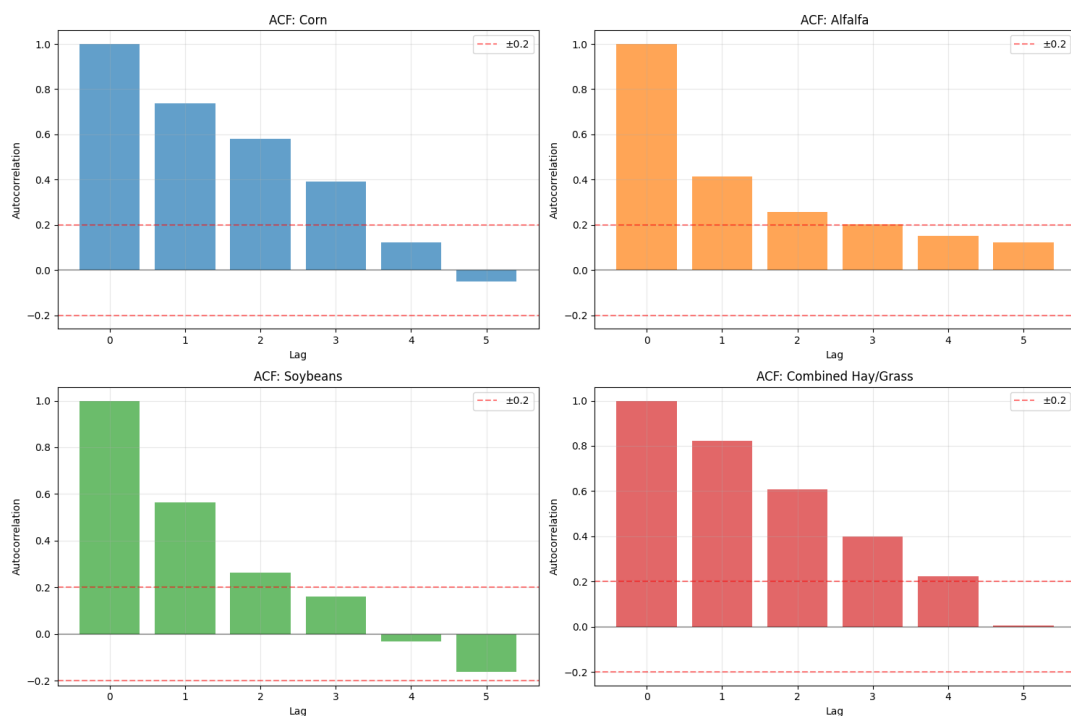
We observed a class imbalance issue for the plant strategy in the CSB data in terms of the total acreage, but the class is balanced in polygon level



Data - Autocorrelation

- Acreage exhibited the autocorrelation according to the ACF plot

Acreage per crop type



Data - Weather Data

-a panel of county-level weekly weather observations, containing detailed temperature, precipitation, growing degree days, and aridity measures along with their deviations from historical norms and prior periods.

variables:

- **avgt** (*Average temperature*) — Mean air temperature over the specified period (°F).
- **mint** (*Minimum temperature*) — Lowest recorded air temperature during the period (°F).
- **totr** (*Total precipitation*) — Cumulative precipitation (rain/snow equivalent) over the period.
- **arid_acc** (*Accumulated aridity index*) — Measure of dryness based on precipitation relative to atmospheric demand; higher values indicate drier conditions.
- **gdd_b10** (*Growing Degree Days, base 10°F*) — Heat accumulation above 10°F used to estimate crop development.
- **gdd_b44** (*Growing Degree Days, base 44°F*) — Heat accumulation above 44°F used for modeling crop growth stages.



Data - Soil Data

USDA gSSURGO provides detailed soil properties, including slope, water table depth, water storage capacity, bedrock depth, and ponding frequency.

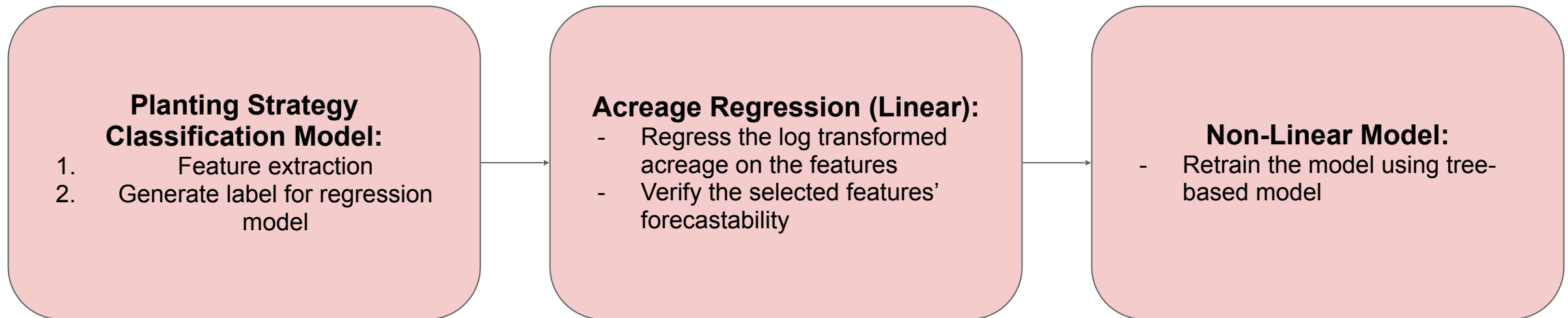
Variables:

- slopegradwta (Average slope (%))
- wtdepannmin (Annual minimum depth to water table)
- wtdepaprjumin (Spring (Apr–Jun) water table depth)
- aws0100wta (Available water (0–100 cm))
- aws050wta (Available water (0–50 cm))
- brockdepmin (Depth to bedrock)
- pondfreqprs (Ponding frequency)



Modeling Methodology

- Goal: Predict the Acreage level with regression model



Feature Selection - Train/test Split

- Timeline split with 2022 as the cutoff,
- Since the data is not i.i.d. we assume the more recent data reflect more validation power
- Training set: 2008-2022 (~88%)
- Validation set: 2023-2024 (~12%)

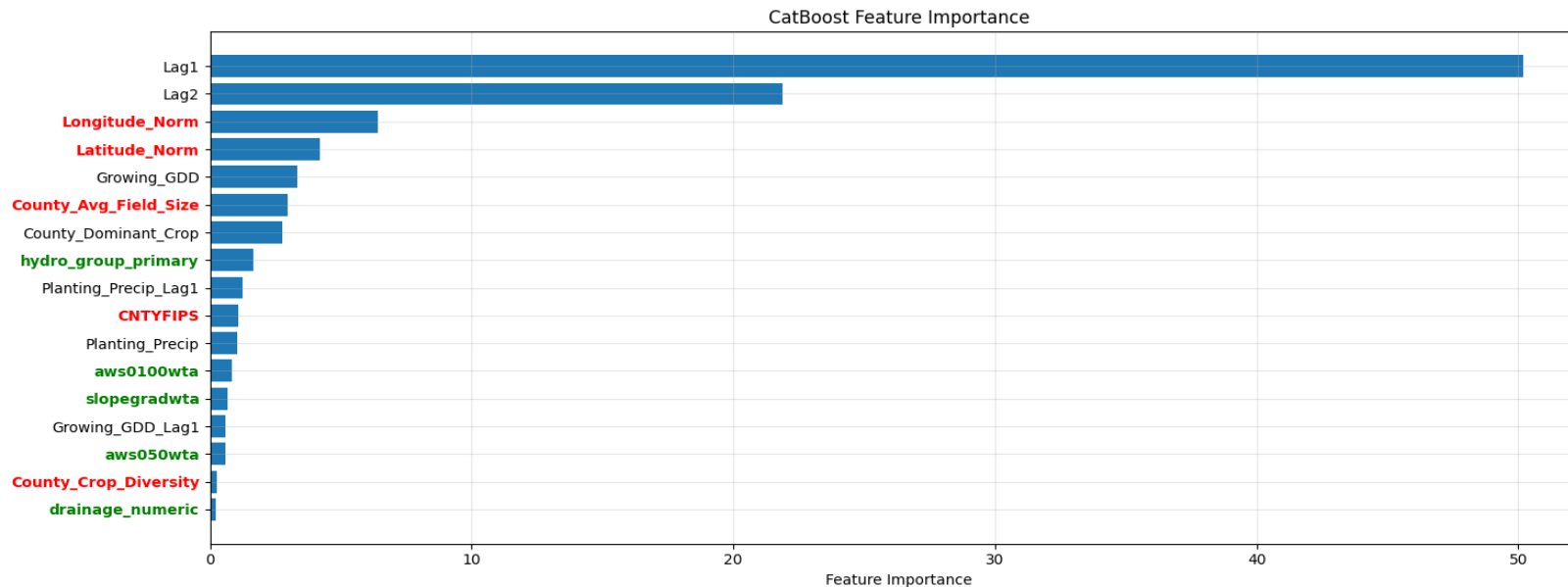


Classification Model - Catboost

Goal: 1) Generate polygon level label 2) feature selection

We selected CatBoosting for three advantages:

1. Robust accuracy (vs KNN)
2. Training efficiency (vs LSTM)
3. Model's built-in feature importance served as the feature selection



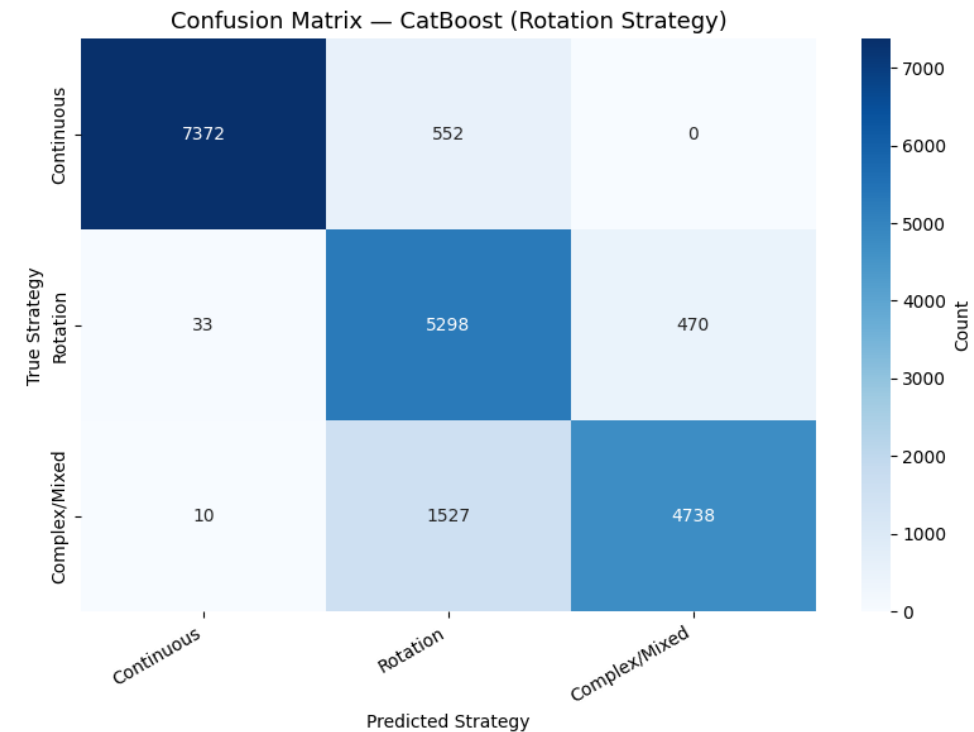
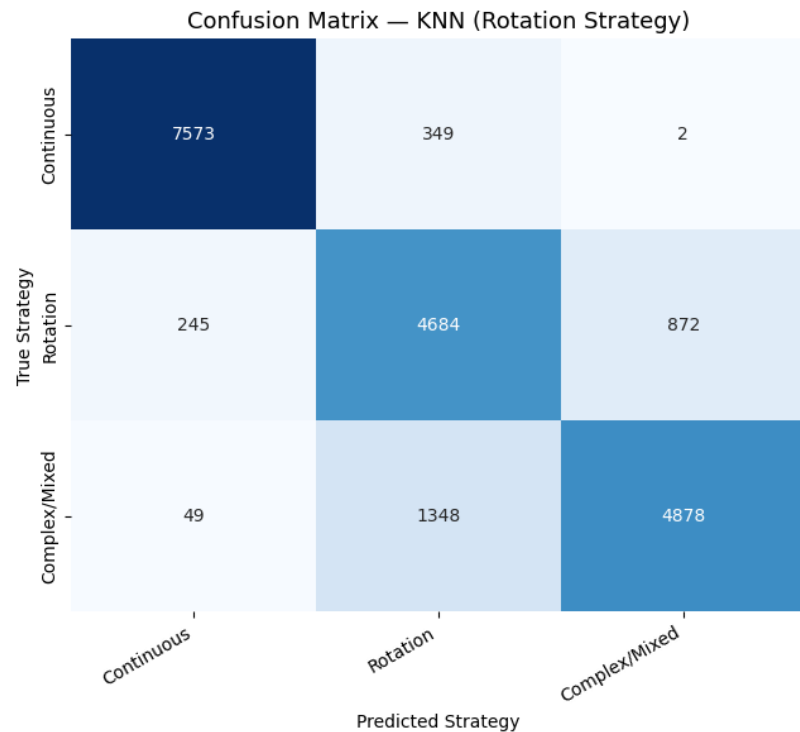
Classification Model - Catboost VS KNN

We noticed Catboost is slightly better than KNN
(Confusion matrix)

True Continuous: F1 (~96.12% vs ~95.92%)

True Rotation: F1 (~80.41% vs ~76.90%)

True Complex/Mixed: F1 (~82.52% vs ~81.12%)



Classification Model - Catboost VS KNN

Goal: 1) Generate polygon level label 2) feature selection

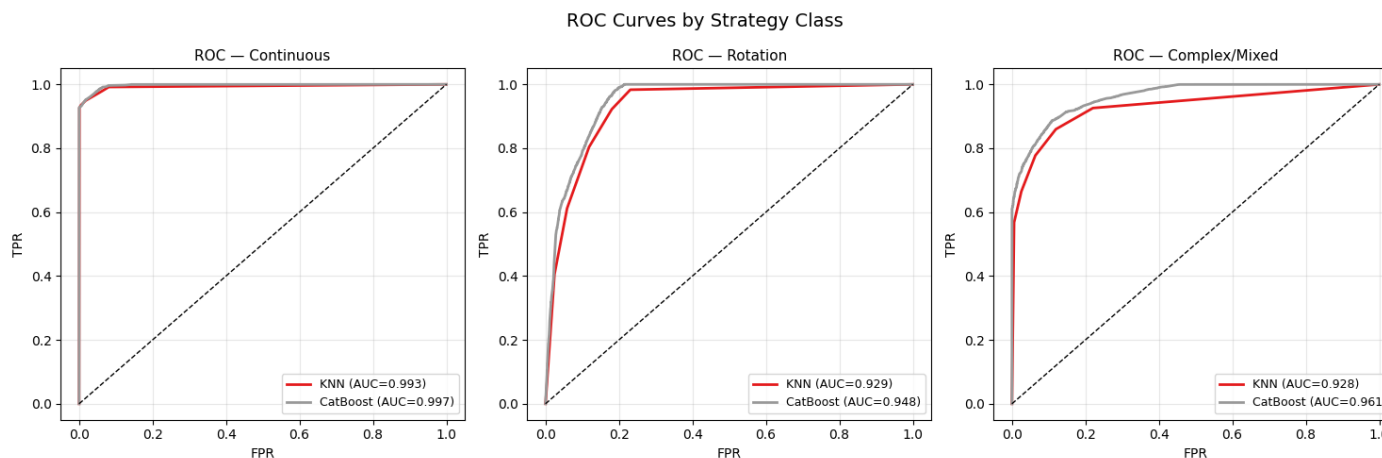
accuracy (~87.0% vs ~85.7%)

precision — macro (~87.40% vs ~84.82%)

recall — macro (~86.62% vs ~84.68%)

F1 — macro (~86.35% vs ~84.64%)

ROC AUC — macro (~96.87% vs ~94.99%)



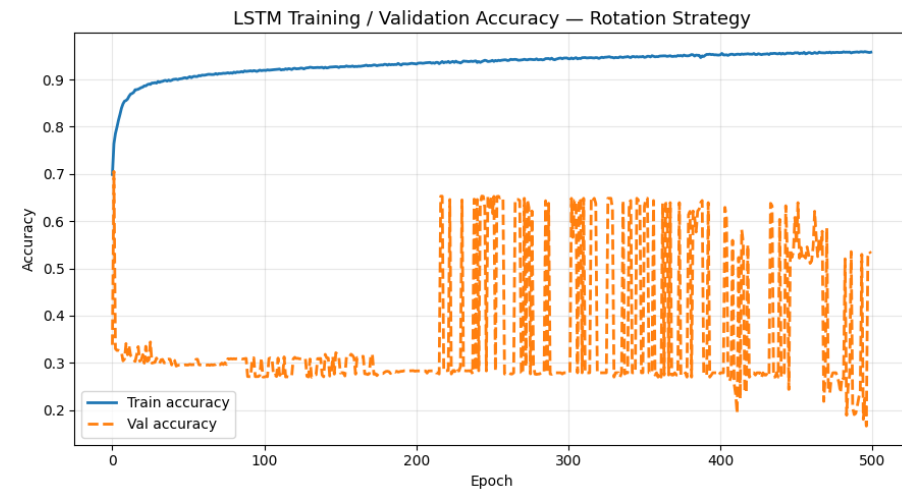
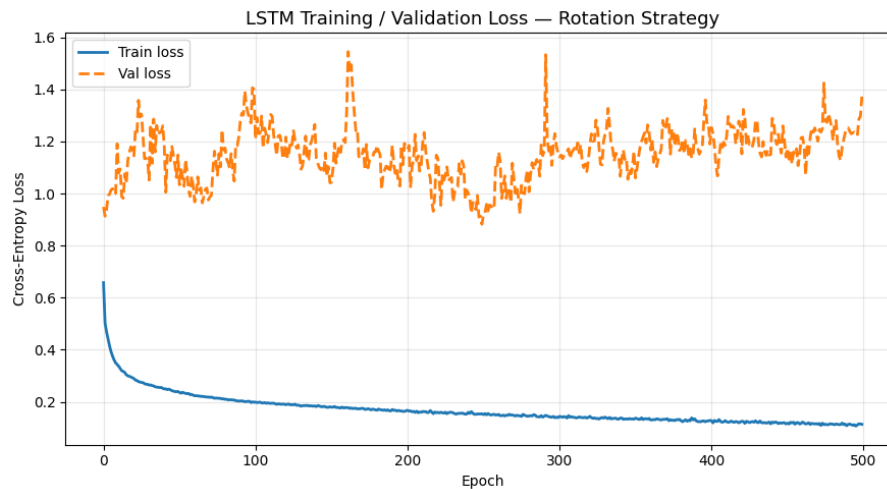
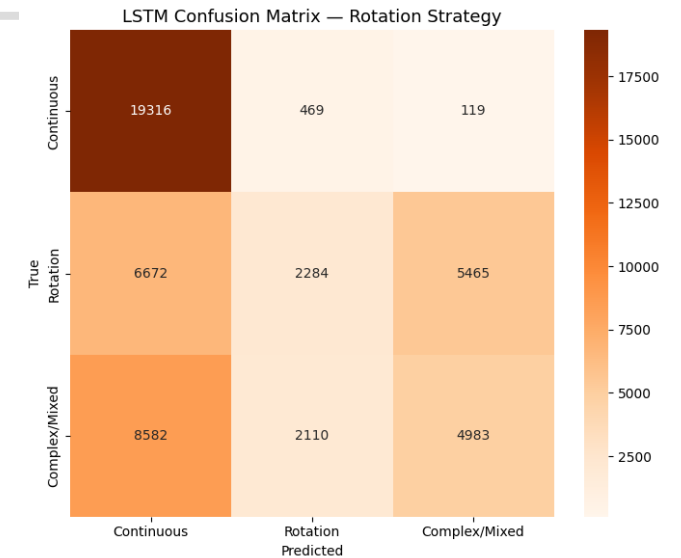
	KNN			
	precision	recall	f1-score	support
Continuous	0.9626	0.9557	0.9592	7924
Rotation	0.7341	0.8074	0.7690	5801
Complex/Mixed	0.8481	0.7774	0.8112	6275
accuracy			0.8568	20000
macro avg	0.8482	0.8468	0.8464	20000
weighted avg	0.8604	0.8568	0.8576	20000

	CatBoost			
	precision	recall	f1-score	support
Continuous	0.9942	0.9303	0.9612	7924
Rotation	0.7182	0.9133	0.8041	5801
Complex/Mixed	0.9098	0.7551	0.8252	6275
accuracy			0.8704	20000
macro avg	0.8740	0.8662	0.8635	20000
weighted avg	0.8876	0.8704	0.8730	20000



Classification Model - Deep Learning

- We trained LSTM model for classification
- LSTM specification:
 - Hidden layer = 2 (since simple model yields good result)
 - # Epoch: 500 (due to computational constraints)
 - Sample size: 50000 (cpu crashed beyond this level)
 - Batch size: 64; Optimizer: Adam
 - Loss: Entropy; drop_out: False
- Lstm tends to memorize instead of generalize
- We prefer simple tree-based model over a deep model



Baseline MLR Model

Goal: Build an interpretable baseline model for crop acreage prediction

Response variable:

$\log(1 + \text{acreage_area_proxy})$

Predictors included:

- Crop type and year
- Soil features
- Weather variables
- County-level rotation strategy feature

Predictor <chr>	Meaning <chr>
crop_code	Crop type
year	Year
Shape_Area_scaled	Spatial polygon area
slopegradwta_scaled	Weighted average slope
brockdepmin_scaled	Minimum bedrock depth
wtdepannmin_scaled	Annual minimum water table depth
wtdeparjunmin_scaled	April-June minimum water table depth
pondfreqprs_scaled	Ponding frequency
aws050wta_scaled	Available water storage, 0-50 cm
aws0100wta_scaled	Available water storage, 0-100 cm



Model Performance

Metric	Value
RMSE	1.574
MAE	1.218
R ²	0.653
Adjusted R ²	0.649
MSE on original acreage scale	1.24×10^{15}

Interpretation:

The model explains about **65%** of the variation in log-transformed acreage. The log-scale RMSE is **1.574**. The original-scale MSE is large because acreage values are large and squared errors amplify large prediction gaps.



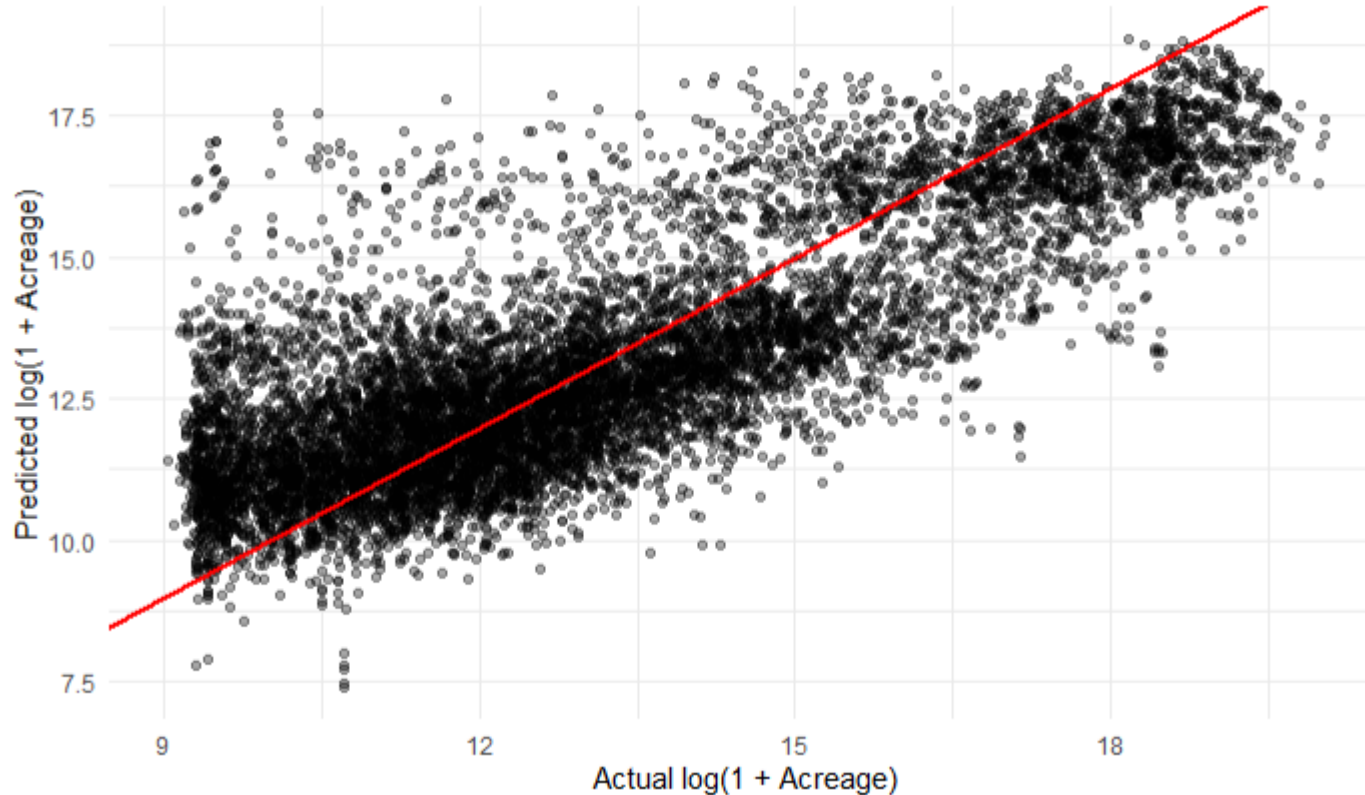
Actual vs Predicted Log Acreage

Main observations:

- Predictions follow the general increasing trend.
- Many points are close to the 45-degree reference line.

The spread is still large, especially in middle and high acreage ranges.

Actual vs Predicted Log Acreage: Full MLR Model



Interpretation:

MLR captures broad acreage patterns but misses some detailed variation

Model Diagnostics

Residual diagnostics show model limitations:

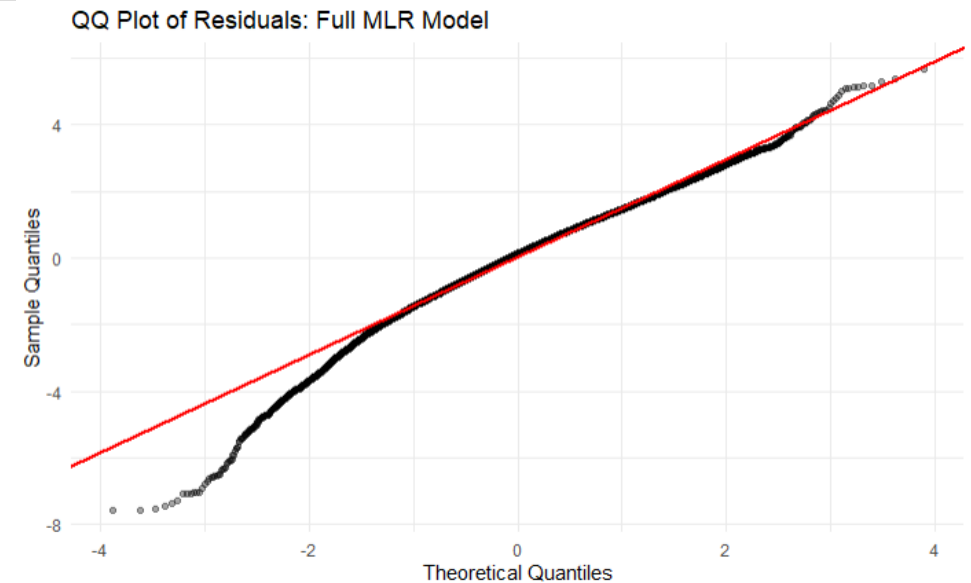
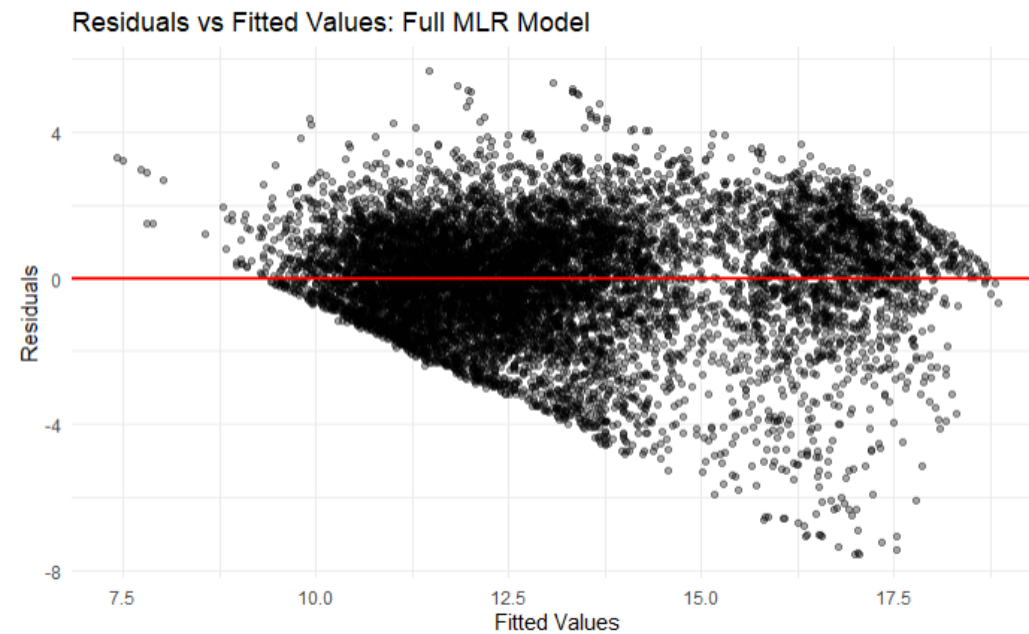
- Residuals are not randomly scattered.
- Error spread increases at higher fitted values.
- QQ plot shows tail deviations.
- Some extreme acreage cases are not well captured.

Implication:

Heteroskedasticity.

The model overpredicts acreage for some observations.

These diagnostics suggest that crop acreage is likely affected by nonlinear relationships or interaction effects that a standard linear model cannot fully capture.



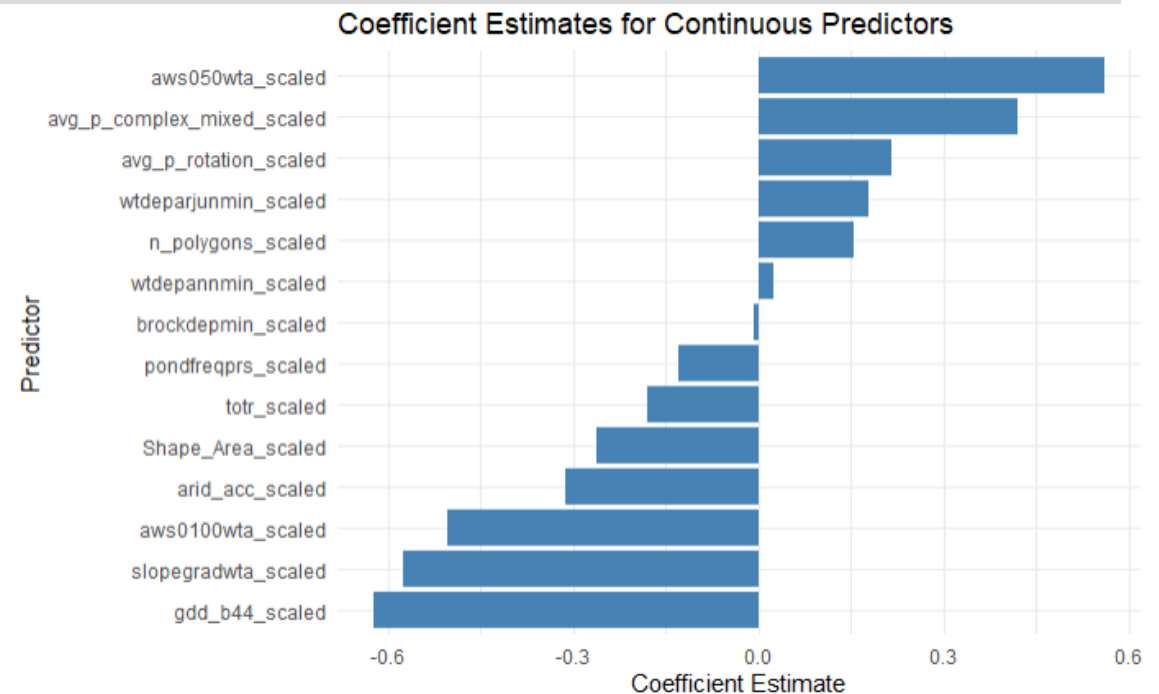
Coefficient Insights

Stronger positive associations:

- aws050wta (available water storage in the top 50 cm of soil)
- avg_p_complex_mixed (average probability of

Stronger negative associations:

- gdd_b44 (growing degree days above base 44
- slopegradwta (weighted average land slope)
- arid_acc (aridity accumulation index)



Takeaway from Initial MLR

What this baseline tells us:

1. Soil, weather, crop, and rotation features explain meaningful acreage variation.
 2. Rotation strategy features add useful management-pattern information.
 3. MLR is interpretable but limited.
 4. Residual patterns suggest nonlinear or interaction effects.
-

Next step:

This motivates the next modeling stage, where more flexible models can better capture nonlinear relationships and interaction effects.



Random Forest Regression

- **Preprocess data:** Load merged dataset, drop missing values, define predictors (exclude target and IDs)
- **Prepare per crop:** Filter by crop, log-transform acreage, split into train/test (80/20)
- **Train model:** Fit RandomForestRegressor (100 trees) for each crop
- **Evaluate:** Compute test R-squared and RMSE
- **Interpret & output:** Extract feature importance, rank models, and save results



Predictive accuracy

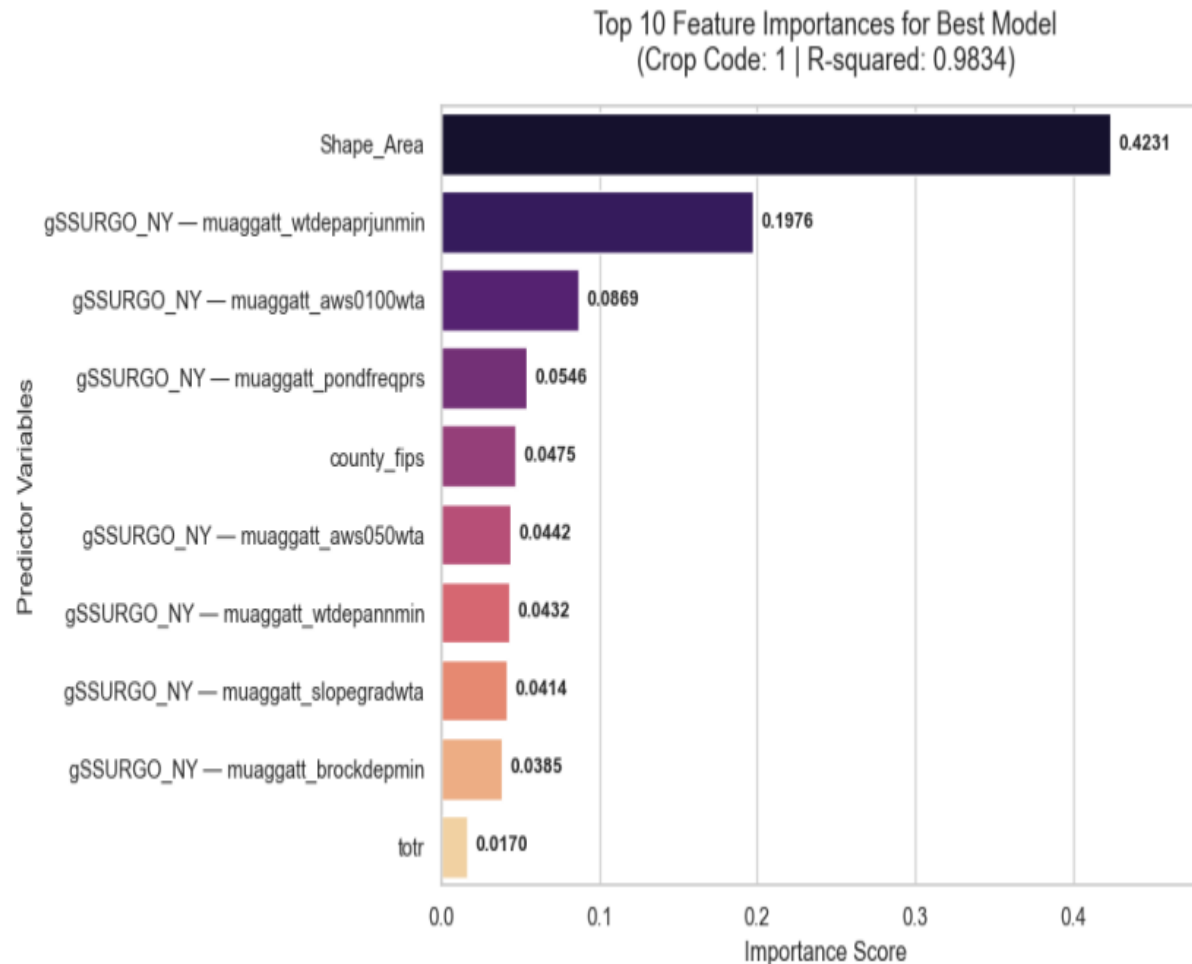
Very high predictive performance across crops: The top Random Forest models achieve test R-squared values between 0.82 and 0.98, meaning the predictors explain 82%–98% of the variation in crop acreage. For example, Corn has test R-squared=0.9834, indicating extremely accurate

--- Top 10 Models by Test R-squared ---

Crop Name: Corn	Test R-squared: 0.9834	RMSE: 0.3161
Crop Name: Apples	Test R-squared: 0.9531	RMSE: 0.6337
Crop Name: Soybeans	Test R-squared: 0.9438	RMSE: 0.4234
Crop Name: Other Hay / Non-Alfalfa	Test R-squared: 0.9197	RMSE: 0.6224
Crop Name: Grassland/Pasture	Test R-squared: 0.9149	RMSE: 0.6034
Crop Name: Alfalfa	Test R-squared: 0.8837	RMSE: 0.6449
Crop Name: Cabbage	Test R-squared: 0.8676	RMSE: 0.7689
Crop Name: Peas	Test R-squared: 0.8405	RMSE: 0.6622
Crop Name: Evergreen Forest	Test R-squared: 0.8317	RMSE: 0.4538
Crop Name: Grapes	Test R-squared: 0.8215	RMSE: 1.0668



Feature Importance



- **Soil and geographic factors are the most influential predictors:** For the best-performing model (Crop Code 1), the most important variable is Shape_Area (importance ≈ 0.423), followed by several soil characteristics such as water table depth, available water storage, and slope. These results suggest that land characteristics strongly influence crop acreage allocation.
- **Weather variables play a smaller role in this model:** Although weather variables were included in the predictors, their importance scores are relatively lower (e.g., totr ≈ 0.017). This suggests that soil properties and land area may be more important determinants of acreage for this crop than short-term weather conditions.

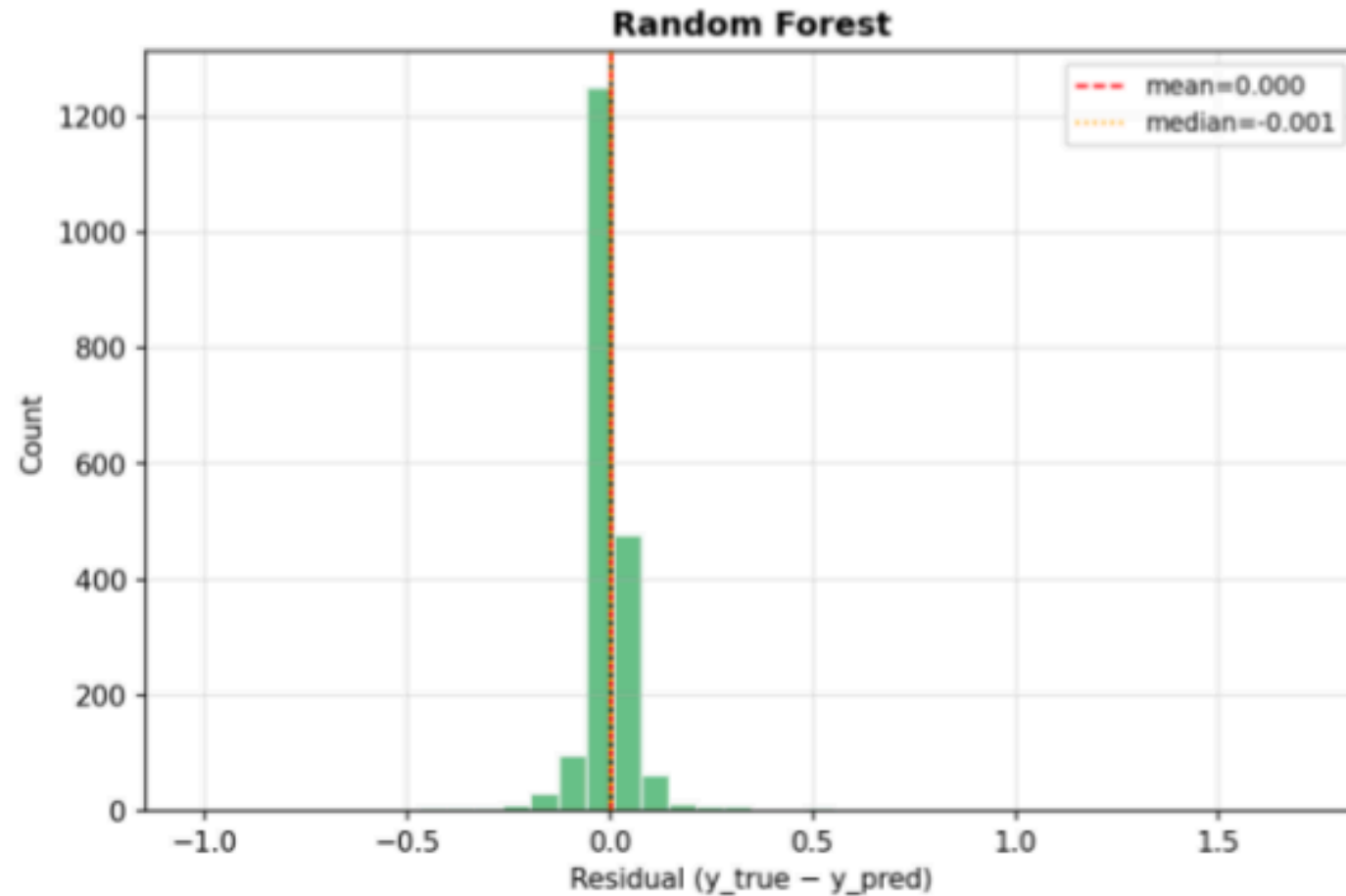


Uncertainty Quantification

- **Method:** Use **out-of-bag (OOB) residuals** from Random Forest to estimate uncertainty without retraining
- **Residual distribution:** Compute empirical distribution of OOB residuals (prediction errors)
- **Prediction intervals:** Construct intervals by adding/subtracting residual quantiles (e.g., 2.5% and 97.5%) to point predictions
- **Evaluation metrics:** Assess uncertainty using **coverage** ($\approx 95\%$ target) and **interval width**
- **Outcome:** Produces well-calibrated, low-width prediction intervals with near-nominal coverage

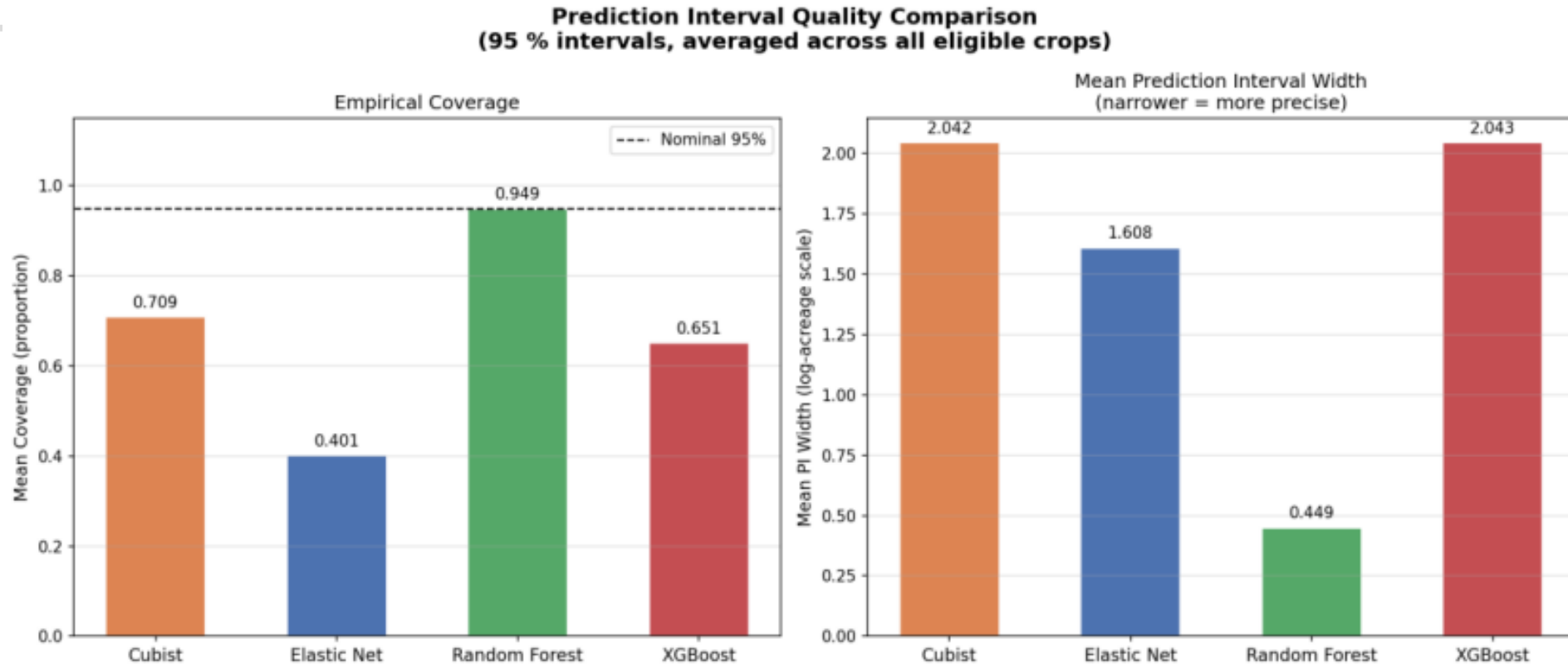


Uncertainty Quantification



The residual distributions clearly show that Random Forest provides unbiased, low-variance predictions.

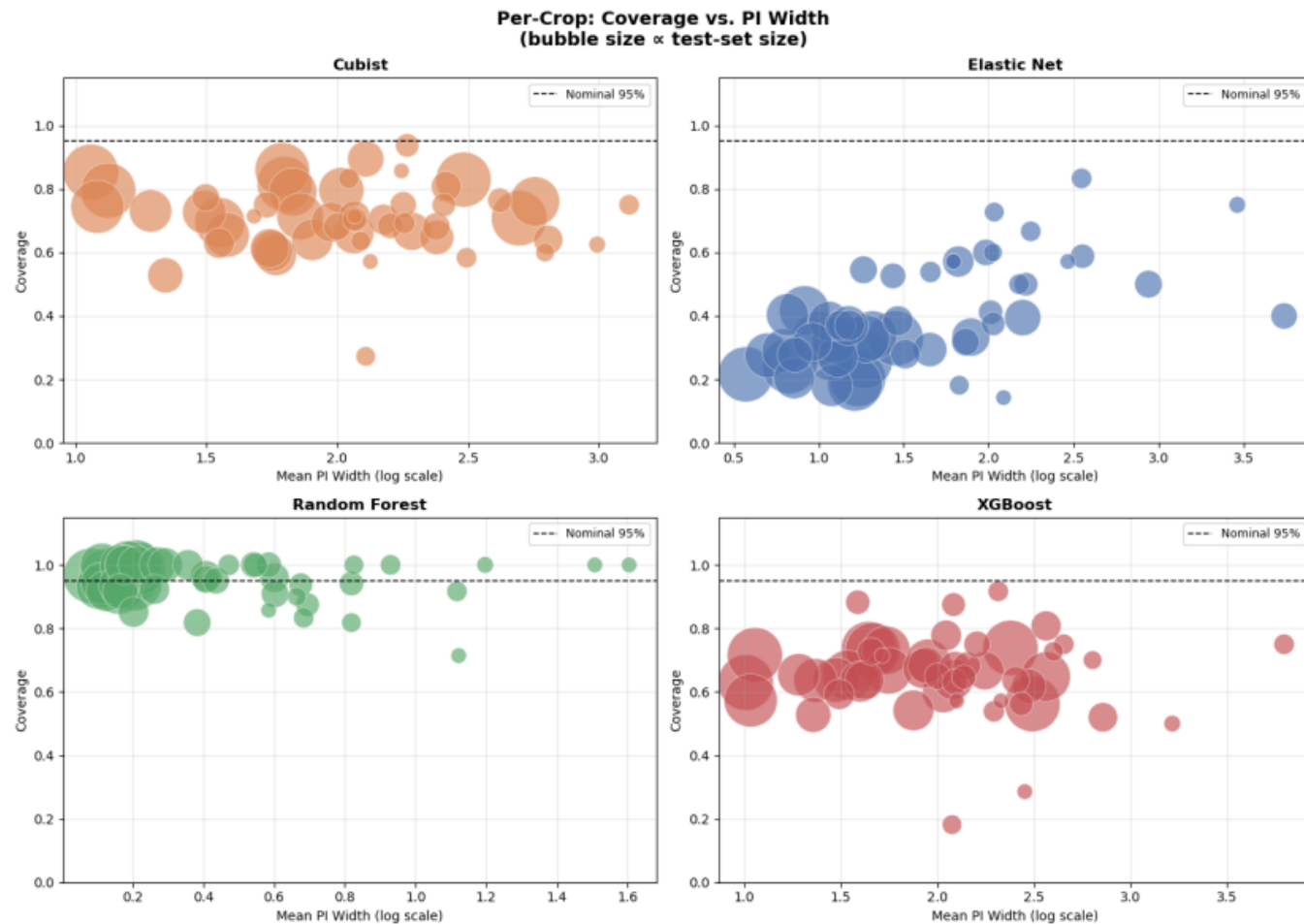
Random Forest Result



Random Forest provides well-calibrated prediction intervals with near-nominal coverage (94.9%) and the narrowest widths, indicating both reliability and precision.



Random Forest Result



Random Forest consistently delivers well-calibrated and precise prediction intervals across all crops.



Summary

Random Forest regression achieves **high predictive accuracy and low uncertainty** by capturing nonlinear patterns through ensemble learning, yielding strong out-of-sample performance (high test R-squared, low RMSE). Its use of OOB residuals enables efficient construction of prediction intervals with **near-nominal coverage and narrow widths**, indicating precise and well-calibrated uncertainty estimates.



Key Findings

- **Pipeline:** Multi-stage feature engineering integrating crop sequence history, weather, soil and spatial data
- **Key insight:** Agriculture is highly autocorrelated and environmentally dependent
- **Crop classification:** CatBoost chosen for interpretability (87% accuracy); LSTM achieves 91.92% but at cost of transparency
- **Acreage Estimation:** Baseline MLR explains ~65% of variance, motivating the shift to nonlinear models. Random Forest achieves R^2 up to 0.98, providing unbiased low-variance predictions



Area For Improvement

Data Limitations

- CSB annual data smooths out monthly seasonality
- County-level aggregation loses field-level granularity

Model Limitations

- Deep learning limited to 50K samples (full dataset: 5.9M)
- CNN/RNN struggles with crop type prediction
- Random Forest trained independently per crop type

Future Directions

- Satellite imagery & Vegetation Index
- Transformer/GNN architectures
- Unified multi-output model for cross-crop dependencies
- Conformal prediction for uncertainty quantification



Acknowledgements/Thanks

We would like to thank our advisors for their guidance and support throughout this project:

- **Luca Sartore, Ph.D.** NISS Contractor (Senior Research Fellow), U.S. Department of Agriculture
- **David Ruppert, Ph.D.** Andrew Schultz Jr. Professor of Engineering and Professor of Statistics and Data Science at Cornell University



Appendix

reference:

- <https://www.osc.ny.gov/reports/new-york-farms-and-farmland-decrease-higher-rate-nationally-2025>
- <https://www.dtnpf.com/agriculture/web/ag/crops/article/2023/07/31/farmers-know-production-costs-higher>

